Wilson Sauthoff (he/him)

Ph.D. Candidate, Colorado School of Mines, Hydrologic Sciences & Engineering Program

---

Contact Information

School of Earth, Space and Planets
Department of Geophysics
Colorado School of Mines
1500 Illinois St.
Golden, CO 80401 USA

https://wsauthoff.github.io
https://glaciology.mines.edu

Education

Hydrologic Science & Engineering Ph.D. (in progress, expected 12/2026)
Data Science M.Sc., 2025
Advisor: Dr. Matthew Siegfried
Colorado School of Mines, (8/2020–Present)

Ocean Sciences M.Sc., 2016
Advisors: Drs. Christina Ravelo & Matthew McCarthy
University of California Santa Cruz, (9/2013–9/2016)

Environmental Science and Management B.Sc., Oceanography minor, 2013
Advisor: Dr. Tessa Hill
University of California Davis, (9/2010–6/2013)

Technical Proficiencies

Languages: Python (proficient), R, SQL, Bash, MATLAB, LaTeX
Machine Learning: scikit-learn, XGBoost, positive-unlabeled learning, ensemble/bagging methods, regularized regression, tree-based models, cross-validation design, class imbalance
Data & Infrastructure: pandas, NumPy, xarray, Dask, GDAL, Git/GitHub/GitLab, CI/CD, Docker, AWS S3, cloud/JupyterHub compute, cloud-optimized formats including GeoTIFF (COG)
Remote Sensing: infrared (IR) satellite imagery, satellite altimetry (ICESat-2, CryoSat-2), multi-sensor fusion, gridded derived products, time-series change detection, ice-penetrating radar, GIS/GPS
Statistics: hypothesis testing, regression modeling, time-series trend and anomaly analysis, spatial statistics, uncertainty quantification, resampling and bootstrap methods, model evaluation

Professional Experience

Lockheed Martin Space Littleton, Colorado
Machine Learning Operations (MLOps) Engineering Intern, Validation of ML models for climate intelligence products; Continuous integration and continuous delivery (CI/CD) of product updates to meet customer needs, (5/2023–5/2026)

National Science Foundation Office of Polar Programs Alexandria, Virginia
Science Assistant, Facilitated merit-based review of scientific proposals for funding consideration; Performed proposal and award data analysis to assess trends in metrics including institution parameters, proposing researcher demographics, etc. to achieve portfolio balance and broader impacts; Wrote internal documentation surrounding funding decisions; Project management of special initiatives to increase stakeholder engagement; Executive Secretary to Executive Management Board of cross-agency management partners of the U.S. Antarctic Program, (3/2018–8/2020)

Pacific Architects & Engineers McMurdo Station, Antarctica

Various roles, Coordinated site assessments, remediation, monitoring, and training activities at primary logistical hub and remote field camps; Ensured compliance with regulatory policies pertaining to environmental protection and waste management observing the Antarctic Treaty; Collected and analyzed environmental samples; Maintained GIS and GPS records; Authored technical reports (7/2017–2/2018); Ensured timely, safe, and effective passenger and courier movements at remote research station and airfields; Dispatched passenger and crew ground movements, (10/2016–2/2017)

Monterey Bay Aquarium Research Institute Moss Landing, California

Ocean Chemistry Intern, Researched deep-sea oil leaks using laser Raman spectroscopy to model environmental fate of oil pollution and methane seeps; Coded data reduction program; Mobilized materials for fieldwork, (6–9/2013)

Sacramento Air Quality Management District Sacramento, California

Mobile Sources Intern, Implemented incentive program oversight; Analyzed air quality surveys; Prepared technical documents including grant final reports; Organized public outreach efforts, (6/2012–6/2013)

Field Experience

Mines Geophysics Field Camp Steamboat Springs, CO

Student, Transient electromagnetics, ground-penetrating radar, distributed acoustic sensing, near-surface seismic, reflection seismic, gravity, and GPS methods, (5/2024)

Peer-Reviewed Publications

Ross, N., Sanderson, R. J., Kulessa, B., Siegert, M., Paxman, G. J. G., Nichols, K., Siegfried, M. R., Jamieson, S. S. R., Bentley, M. J., Jordan, T. A., Batchelor, C. L., Small, D., Eisen, O., Winter, K., Bingham, R. G., Callard, S. L., Carr, R., Dow, C., Fricker, H. A., Hill, E., Hills, B. H., Hofstede, C., Jeofry, H., Napoleoni, F., & **Sauthoff, W.** (2026) Review Article: The Foundation-Patuxent-Academy ice stream system, Antarctica. *The Cryosphere.* https://doi.org/10.5194/tc-20-3705-2026

Smith, B., Sutterley, T., Fricker, H., Padman, L., Siegfried, M., Black, T., Felikson, D., Freer, B., Gibbons, A., Jelley, B., King, M., Medley, B., Morlighem, M., Sadlik, C., **Sauthoff, W.**, & Nuemann, T. (2026) ICESat-2 land ice products resolve Greenland and Antarctic ice-sheet height changes on seasonal to multiyear time scales. *Journal of Glaciology*. https://doi.org/10.1017/jog.2026.10152

Snow, T., Holdgraf, C., **Sauthoff, W.**, Scheick, J., Abrahams, E., Millstein, J., Bhangar, S., Boettiger, C., Colliander, J., Lopez Espinosa, L. A., Holmes, E. E., Kennedy, J. H., Lowndes, J. S., Mandel, A. I., Panda, Y., Perez, F., Swinski, J.-P., Teucher, A., Siegfried, M. R. (2026) A path to better science through co-creation and open infrastructure. *AGU Perspectives of Earth and Space Scientists.* https://doi.org/10.1029/2025CN000295

Sauthoff, W., Siegfried, M., Venturelli, R., & Smith, B. (2026) Dynamic boundaries of Antarctic active subglacial lakes reveal underestimated water volumes and overestimated lakebed active area. *Geophysical Research Letters. 53,* e2025GL117121. https://doi.org/10.1029/2025GL117121

Horgan, H., Stewart, C., Stevens, C. Dunbar, G., Balfoort, L., Schmidt, B., Washam, P., Werder, M., Mandeno, D., Marschalek, J., Hulbe, C., Holschuh, N., Levy, R., Hurwitz, B., Johnson, K., Lawrence, J., Morgenstern, Mullen, A., **Sauthoff, W.**, Siegfried, M., Still, H., van der Flierdt, T., Venturelli, R., Whiteford, A. (2025) A West Antarctic grounding-zone environment shaped by episodic water flow. *Nature Geoscience*. https://doi.org/10.1038/s41561-025-01687-3

Verboncoeur, H., Siegfried, M., Holschuh, N., Winberry, J., Byrne, D., **Sauthoff, W.,** Sutterley, T., Medley, B. (2025) Multi-decadal evolution of Crary Ice Rise region, West Antarctica, amid modern ice-stream deceleration. *Journal of Glaciology*. https://doi.org/10.1017/jog.2024.79

Freer, B. I. D., Marsh, O. J., Fricker, H. A., Hogg, A. E., Siegfried, M. R., Floricioiu, D., **Sauthoff, W.**, Rigby, R., Wilson, S. F. (2024) Coincident Lake Drainage and Grounding Line Retreat at Engelhardt Subglacial Lake, West Antarctica. *Journal of Geophysical Research: Earth Surface*. https://doi.org/10.1029/2024JF007724

Stubblefield, A., Meyer, C., Siegfried, M., **Sauthoff, W.**, & Spiegelman, M. (2023) Reconstructing subglacial lake activity with an altimetry-based inverse method. *Journal of Glaciology*. https://doi.org/10.1017/jog.2023.90

Brault, E., Koch, P., McMahon, K., Broach, K., Rosenfield, A., **Sauthoff, W.**, Loeb, V., Arrigo, K., & Smith, W. (2018). Carbon and nitrogen isoscapes in West Antarctica reflect oceanographic transitions. *Marine Ecology Press Series*. https://doi.org/10.3354/meps12524

Serrato Marks, G., LaVigne, M., Hill, T., **Sauthoff, W.**, Guilderson, T., Roark, B., Dunbar, R., & Horner, T. (2017). Reproducible Ba/Ca variations recorded by northeast Pacific bamboo corals. *Paleoceanography and Paleoclimatology*. https://doi.org/10.1002/2017PA003178

Moffitt, S., Moffitt, R., **Sauthoff, W.**, Hewett, K., Davis, K., & Hill, T. (2015) Paleoceanographic insights on recent oxygen minimum zone expansion: Lessons for modern oceanography. *PLOS ONE*. https://doi.org/10.1371/journal.pone.0115246

Other publications

Sauthoff, W., Snow, T., Millstein, J. D., Colliander, J., & Siegfried, M. R. (2024), Democratizing science in the cloud, *Eos*, 105, https://doi.org/10.1029/2024EO240385. Published on 30 August 2024.

Sauthoff, W., Siegfried, M. R., & Smith, B. E. (2024). Evolving Outlines of Antarctic Active Subglacial Lakes Using An Image Processing Algorithm On Gridded Altimetry Data. In IGARSS 2024 - 2024 IEEE International Geoscience and Remote Sensing Symposium (pp. 8687–8691). https://doi.org/10.1109/IGARSS53475.2024.10642198

Snow, T., Millstein, J., **Sauthoff, W.**, Scheick, J., & Siegfried, M. (2023). CryoCloud response to NASA NNH23ZDA005L "Request for Information: Scientific Data and Computing Architecture to Support Open Science". Zenodo. https://doi.org/10.5281/zenodo.7662908

Snow, T., Millstein, J., Scheick, J., **Sauthoff, W.**, & others. (2023). CryoCloud JupyterBook (2023.01.26). Zenodo. https://doi.org/10.5281/zenodo.7576602

Siegfried, M., Schroeder, D., **Sauthoff, W.**, Smith, B. (2021) Investigating a large subglacial lake drainage in East Antarctica with ice-penetrating radar. *SEG Technical Program Expanded Abstracts*. https://doi.org/10.1190/segam2021-3582777.1

Ruck, K., Sheffield Guy, L., Chetty, S., French, N., Seefeldt, M., Carpenter, P., Frey, K., Griffith, P., Prior-Jones, M., **Sauthoff, W.** (Eds.). (2020). Polar Technology Conference 2020 Report. Arctic Research Consortium of the US (ARCUS). https://www.arcus.org/publications/31301

Isern, A., Burtler, N., Shanhun, F., **Sauthoff, W.** (Eds.) (2019). Report on International Ross Sea Region Collaboration and Coordination Workshop. Scientific Committee on Antarctic Research. https://www.scar.org/scar-news/pais-news/ross-sea-workshop-report/

Sauthoff, W., Ravelo, C., McCarthy, M. (2016). Nitrogen isotopes of amino acids in marine sediment: A burgeoning tool to assess organic matter quality and changes in supplied nitrate $\delta^{15}N$. M.Sc. thesis, University of California Santa Cruz. http://escholarship.org/uc/item/63r5v4zj

Sauthoff, W. (2013) Gulf of Alaska and California bamboo corals: A new archive of climate change. *Explorations: The UC Davis Undergraduate Research Journal*, Vol. 15. http://explorations.ucdavis.edu/2013/sauthoff_wilson.html

Data Sets	**Sauthoff, W.** & Siegfried, M. (2025) Antarctic subglacial hydropotential. *Zenodo*. https://doi.org/10.5281/zenodo.16243278
	Smith, B., & **Sauthoff, W.** (2025) CryoSat-2 SARIn Height Change and Reference DEM for Antarctica. *Zenodo*. https://doi.org/10.5281/zenodo.14963551
	Sauthoff, W., & Freer, B. (2024) Subglacial water flow paths beneath Whillans Ice Plain, West Antarctica. *UK Polar Data Centre*. https://doi.org/10.5285/0df5d4e9-2fcd-4420-b403-24d76848a5a5
Code and software	**Sauthoff, W.**, & Siegfried, M. (2025) wsauthoff/Sauthoff-2025-GRL code repository. https://doi.org/10.5281/zenodo.15758712
	Sauthoff, W., & Siegfried, M. (2023) wsauthoff/wsauthoff/contours2polygons. https://zenodo.org/records/7725122
Educational content	**Sauthoff, W.**, Siegfried, M. (2025) Streaming cloud-hosted ICESat-2 ATL15 (Gridded Antarctic/Arctic Land Ice Height) to calculate dh/dt trends. CryoCloud Jupyterbok tutorial. https://book.cryointhecloud.com/icesat2-atl15-dhdt
	Sauthoff, W., Lopez, L. Scheick, J., Snow, T. (2023) Using ICESat-2 ATL15 (Gridded Arctic Land Ice Height) to investigate ice-surface height anomalies. CryoCloud Jupyterbok tutorial. https://book.cryointhecloud.com/is2-atl15-surface-height-anomalies
Selected Conference Presentations and Service	Student convener, Open Source Geospatial Workflows in the Cloud: Tools and Techniques for Data Access, Analysis, Visualization, Storytelling, and Sharing in the Python and Jupyter ecosystem, 2025 AGU Fall Meeting, New Orleans, La. (12/2025)
	Primary convener, Accelerating Scientific Discovery and Interdisciplinary Collaboration through Cloud Computing Hubs and Tools, 2024 AGU Fall Meeting Session U001, Washington, D.C. (12/2024)
	Evolving outlines of Antarctic active subglacial lakes using an image processing algorithm on gridded altimetry data (talk), 2024 IEEE International Geoscience and Remote Sensing Symposium Athens, Greece (7/2024)
	Primary convener, Advancing Cryospheric Research and Community Engagement Through Open-Science Tools, Data, and Collaboration, AGU Fall Meeting, San Francisco, Calif. (12/2023)
	Altimetry-based, surface-deformation delineation algorithm reveals tens of new active subglacial lake candidates across Antarctica (talk), 2023 AGU Fall Meeting Session C14A-04, San Francisco, Calif. (12/2023)
	Surface-deformation delineation algorithm reveals subglacial lake candidates and underestimates of subglacial volume fluxes (poster), 2023 IGS Edges of Glaciology, Limerick, Ireland (7/2023)
	CryoSat-2/ICESat-2 integrated time series and comparison of shoreline evolution in Antarctic active subglacial lakes (poster), 2022 AGU Fall Meeting Session C008, Chicago, Ill. (12/2022)
	ICESat-2-extended time series of subglacial volume fluxes using time-variable shorelines of Antarctic active subglacial lakes (talk), Ice, Cloud and Land Elevation Satellite-2 Science Symposium 2022, Austin, Tex. (10/2022)
	Variable shorelines of Antarctic active subglacial lakes reveal large underestimates of subglacial volume fluxes (poster), 2022 GSA Connects Session T107, Denver, Colo. (10/2022) https://doi.org/10.1130/abs/2022AM-383750
	Refined time series of Antarctic active subglacial lakes: New multimission volume flux estimates using evolving shorelines (talk), 2022 WAIS Workshop, Estes Park, Colo. (9/2022)
	Antarctic active subglacial lake shoreline migration and novel features of ice height anomaly (talk), AGU Frontiers in Hydrology Meeting, San Juan, P.R. (6/2022)

Antarctic active subglacial lake shoreline migration and novel features of ice height anomaly (talk), Spring 2022 Ice, Cloud and land Elevation Satellite-2 Science Team Meeting, Boulder, Colo. (5/2022)

Observing connected subglacial lake drainage at Slessor Glacier, East Antarctica, using ICESat-2 laser altimetry (poster), 2021 WAIS Workshop, Sterling, Va. (9/2021)

Workshop Participation

Participant, Joint IAPSO/IACS Commission on Ice-Ocean Interactions Online workshop, virtual (10/2022)

Participant, CLIVAR Sources, Pathways and ImpaCts of frEsh water in northern and soUthern Polar oceans and seas (SPICE UP), virtual (9/2022)

Co-organizer, International Ross Sea Region Collaboration and Coordination Workshop, Scientific Committee on Antarctic Research (SCAR) XIII International Symposium on Antarctic Earth Sciences (ISAES) Conference, Incheon, Korea (7/2019)

NSF Representative, 2019 Next Generation of Polar Researchers Leadership Symposium, Catalina, CA (5/2019)

Awards

Geophysics Field Camp Fee Scholarship, $350, (6/2024)
National Defense Science & Engineering Graduate Fellowship Honorable Mention, (4/2023)
WAIS Workshop Early Career Researcher Travel Grant, $280, (9/2022)
ESIP Request for Machine Learning Tutorials Grant (Project Partner), $5000, (7-12/2022)
Mines Hydrologic Science & Engineering Student Leadership Award, $400, (7/2022)
WAIS Workshop Early Career Researcher Travel Grant, $651, (9/2021)
Mines Hydrological Science & Engineering Student Leadership Award, $220 (5/2021)
National Science Foundation Antarctic Service Medal, (03/2017)
University of California Santa Cruz Graduate Student Association Travel Grant, $288, (5/2015)
Myers Oceanographic Trust Grant, $1100, (4/2015)
Friends of Long Marine Lab Grant, $950, (1/2015)
PADI Foundation Grant, $1685, (9/2013)
Provost's Undergraduate Fellowship, $894, (12/2011)

Mentoring

Colorado School of Mines
Anastasia Horne, undergraduate and MSc. researcher, (2023-25)
Mia Jungman, undergraduate researcher, (2023)
Venezia Follingstad, undergraduate researcher, (2022), now Univ. of Oregon graduate student
Kayla Hubbard, MSc student, (2021), now National Science Foundation science assistant

University of California Santa Cruz
Natalia Spritzer, Science Internship Program high school intern, (2015)
Sami Chen, undergraduate lab assistant, (2014-2015), now Ph.D. student at Stanford
Nicolette Chiem, undergraduate CalTeach intern, (2014-2015)
Linda Pineda, undergraduate researcher, (2015-2016)

Teaching Experience

Colorado School of Mines
Teaching Assistant, Applications of Remote Sensing (Geophysics 470/570), 1/2022-5/2022

University of California Santa Cruz
Teaching Assistant
Ecology & Society (Environmental Studies 100L), 1/2016-3/2016
Microbiology (Microbiology & Environmental Toxicology 119), 9/2015-12/2015

Our Changing Planet (Oceans 80B), 9/2014–12/2014, 4/2014–6/2014

University of California Davis
Laboratory Teaching Assistant, Mathematics and Science Teaching Program for biological science laboratory (BIS 2A-C lab), 1/2012–3/2012
Biological Sciences Tutor, Student Academic Success Center, 6/2011–12/2011
Instructor, Biological Sciences supplemental course (BIS 98), 4/2011–6/2011